

An evaluation of simulation to support contract costing

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Abstract

As global markets have become more customer oriented, rapid response rates are now often among the most important metrics in business. To achieve the required agility many companies have turned to outsourcing in order to focus on developing their core activities. By its nature, outsourcing increases the complexity of supply chain networks as more companies are drawn into global logistics networks. The relationships between each of these companies are controlled by contractual agreements. The fast pace of modern industry means that these contracts are entered into relatively quickly often without a full understanding of the true cost implications. This paper presents research conducted as part of a project with the aim of developing contract costing software for outsourcing enterprises. Findings are presented from a study conducted on a number of companies in the electronics sector. A simulation-based study focused on one of these companies, with some associated experimentation, is also presented.

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1. Introduction

The practice of outsourcing has been defined as the operation of shifting a transaction previously governed internally to an external supplier through a long-term contract [1] and has been employed as a managerial strategy for hundreds of years. However, over the last 15–20 years outsourcing has received considerable focus and is now seen by many as an astute means of gaining competitive advantage. This can be attributed to trends in the economy that have de-emphasised the benefits of vertical integration (such as economies of scale) and instead focused on the benefits of being specialised and focused (such as rapid response).

This increased attention has made outsourced (or contract) manufacturing one of the fastest growing industries in a broad range of business sectors including electronics, pharmaceuticals, medical devices, automotive, and food and beverage production. Increasingly, firms that traditionally manufactured their own products are outsourcing production and instead focusing on product design, development, and marketing. It has been noted that there are many differences to be considered when outsourcing either manufacturing or services. For instance, when outsourcing a manufacturing function, a host company can precisely define expectations through engineering drawings or standards, whereas this can be far more subjective when dealing with an outsourced service [2].

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Bailey et al. [3] conducted a survey on a broad range of industries in a specific area in Scotland. The largest sector surveyed was the manufacturing industry but responses were also obtained from all major sectors such as finance, education, health and construction. It was found that 70% of the organisations, in the area studied, outsourced at least one activity. In the United States, it has been noted that more than 90% of the major corporations have outsourced some services. Companies engaged in outsourcing were seen to have a percentage growth rate of 9.4%, while that of general industry was at the significantly lower rate of just 7.2%, confirming the strength of this sector [2].

Bendor-Samuel [4] suggests that no organisation, however well managed or well financed, will be able to cope as competitive demands for its cycle time moves from two months to 2 h. The need to improve every aspect of an organisation while increasing focus on its core processes will make outsourcing as foundational to competing successfully in the Internet Age as is access to the Internet itself. He goes on to state unequivocally that “There Is No Alternative”, as outsourcing is the only tool available that provides the ability to improve operations while cutting costs and releasing capital and time that can be used on core areas. While this view may seem extreme, it is clear from the literature that there is a decisive move in industry toward outsourcing, albeit at varying levels. It also worth noting that, unlike many other philosophies, outsourcing cannot be dismissed as nothing more than the latest in a series of short-lived management fashions. For instance, Domberger [5] reports of historical accounts of outsourcing that date back to the eighteenth century.

This paper focuses on the manufacturing perspective of outsourcing and, in particular, on the electronics sector where global outsourcing is set to rise from \$138 billion in 2003 to \$294 billion in 2008 [6]. In this sector the process of inviting vendors to bid for business has become known as the Request For Quotation (RFQ) process. Having decided to outsource a business process (in this study a manufacturing function), a company will issue an RFQ (a document containing a clear description of the work to be completed and outlining all other customer requirements) to each of the contract manufacturers (CMs) deemed relevant to the work. These RFQs often contain elements that directly affect the operational aspects of providing the product/service. However, it can be extremely difficult for a CM to determine their effect on product cost and these aspects are often entirely overlooked. If CM companies are to gain a true understanding of the contracts they enter into, they must adopt a more holistic approach to costing their outsourcing projects. Achieving this requires cost analysis tools of greater sophistication than currently available, so that all elements of the RFQ can be considered.

This study has been conducted as part of a current project named SIMCT (A Simulation based Contract Costing Tool for Outsourcing Enterprises), which is investigating the incorporation of traditional operations research methodologies into the practices of quotation and contract development. The prime objective of this project is to develop software that will help contractors to more accurately cost their responses to incoming RFQs. Focusing primarily on the electronics sector, extensive field research has been conducted in a total of eight companies selected from across the supply chain spectrum. Initial work on this project noted that when responding to such RFQs a CM faces a number of considerations:

- Ensuring that the data received from the customer is complete and representative of the business proposed.
- Ensuring that the cost model is built up correctly and that pricing is in line with the overhead and labour required.
- Ultimately offering a solution or model that is not only better than the customer’s existing model but also more cost effective.

Although this seems logical and straightforward the reality is that this is an extremely difficult task, especially when the time to respond to an RFQ is short. In addition to this, western manufacturing companies are now under increased pressure from companies operating in the cheaper economies of emerging countries such as China and those of Eastern Europe. This emphasis on reducing their prices in order to remain competitive has diminished the CMs’ ability to include safety margins on quoted prices. If the successful CM does not have an accurate understanding of the cost implications of agreeing to the terms of the proposed contract, there are negative repercussions for both parties. The contractor risks operating at a loss while the customer risks poor service levels and the possibility of repeating the time-consuming RFQ process. Further to this, the prevalence of outsourcing has put more importance on formalising the relationships between companies through more stringent contractual agreements. It is therefore in the interest of all involved that the correct outsourcing partners are selected from the beginning.

Building on field study analysis conducted in a range of companies directly involved in outsourcing and on shortcomings identified in existing outsourcing software, this paper establishes the need and justification for developing a simulation based costing tool. As such, the findings of the field study are first presented, followed by an overview

of existing software tools. To evaluate the benefits of using simulation for more accurate contract costing, simulation models were developed for a number of case studies. Presented here, in Sections 4 and 5, are the model and experiments for one of these case study companies.

2. Field study analysis

Over the last number of years a widely recognised categorisation scheme for the various players in supply chain networks has emerged. With this nomenclature, “a company that uses product components from one or more other companies to build a product that it sells under its own company name and brand” [7] is defined as an OEM (original equipment manufacturer). The term CM describes “a third party that performs one or more production operations for a manufacturer who will market the final item under their own name” [8]. CMs can be further categorised depending on their positioning within a supply chain. The phrases “First Tier”, “Second Tier”, etc. are used to denote these positions. Third party logistics provider (3PL) is used to describe businesses that provide one or many of a variety of logistics-related services. These services would include public warehousing, contract warehousing, transportation management, distribution management, and freight consolidation [9]. The term Fourth party logistics provider (4PL) was first coined and registered as a trademark by Accenture, Inc. in 1996 [10]. This trademarked definition describes a 4PL as “an integrator that assembles the resources, capabilities and technology of its own organisation and other organisations to design, build and run comprehensive supply-chain solutions”.

As part of the SIMCT project field research was conducted in companies operating in a variety of supply chain echelons in the electronics sector, more specifically: two OEMs, two first tier CMs, one second tier CM, one 3PL and two 4PLs. This paper investigates the case of a CM responding to an RFQ and will focus primarily on one of the first tier CMs from the field study, described here as *Company X*. *Company X* is a multi-national electronics CM, which provides a range of integrated, value-added services that include design and engineering, multi-layered printed circuit board assembly (PCBA), manufacture of memory modules, optical modules and electronic enclosure systems, cabling/wiring harness assembly, subsystem and system-level assembly and final system test and distribution.

Across the range of companies studied it has been noted that, regardless of the scale of the contract on offer, each company will base an RFQ quote on the sum of the three fundamental costs of material, labour and overhead. The material and labour elements are based on known rates while the overhead figure is often included as a percentage of the quote price. *Company X* has set up a number of centralised cost estimation centres in order to respond to requests for quotations. These centres handle material costs globally and utilise the economy of scale to negotiate with suppliers. The material costs are then returned to the manufacturing facility in question who calculate their in-house labour, operating and overhead costs. All costs are then combined to form the quote that the customer receives. This system means that the true holding cost of material may not be fully understood.

Based on case study information received from a number of the field study companies, it has been noted that the contracts entered into with customers often contain flexibility constraints (see example in Table 1), which allow these customers to increase or decrease the order quantity by an amount depending on the time left to the due date. When the constraints in purchasing agreements with materials suppliers do not mirror these supply constraints a CM can be faced with a number of stock-out or material holding issues.

Invariably, CMs’ purchasing agreements will be much different to those for supply, as there are often minimum purchase requirements that must be satisfied. Also, in the electronics sector many components (such as resistors and capacitors) are purchased on reels that can contain thousands of parts. This does not pose a great problem

Table 1
Flexibility constraints

Weeks from planned delivery	Upside flexibility (%)	Downside flexibility (%)
0–2	15	15
2–4	30	100
4–8	50	100

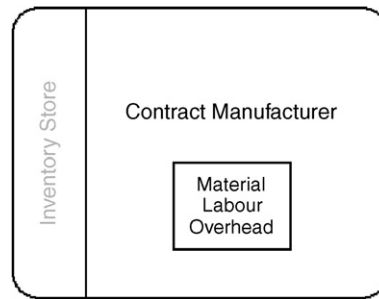


Fig. 1. Current scope of quote estimation.

for high running components that are common to a number of products. However, for a more specific high cost product, this situation can entail a considerable risk, especially if the end product experiences highly fluctuating demand.

In each of the CMs studied it was found that Excel spreadsheets were used for the calculation of its manufacturing costs. *Company X* uses a macro-enabled Excel spreadsheet capable of taking into consideration the varying line configurations available to them. In comparison to the spreadsheet-based aids used in other CMs this is quite a sophisticated tool and experience has shown it to be reasonably accurate although limited with regard to the cost drivers it can capture. When examining the cost of providing goods and services, *Company X* is essentially summing the costs associated with activities conducted within the walls of their manufacturing facility (see Fig. 1) with the cost of materials. By only examining these “4 walls” costs, expenses associated with the supply chain, such as those of logistics are neglected.

The primary finding from this study is that a number of important elements of an RFQ or subsequent contract are not analysed by the companies providing the service or product. The key reason for this is that companies are not aware of any tools in the current market that enables the inclusion of such factors in their cost calculations.

3. Contract costing software requirements

As part of the SIMCT project a number of software categories have been identified as relevant to this field of study [11]. The category deemed most pertinent to the aspects of outsourcing examined here is described as supplier relationship management (SRM). Within the SRM category, the contract management and RFx-based procurement tools are of most relevance (RFx is the collective term for request for information, request for proposal and request for quotation). These tools are used for the sourcing, qualification and selection of suppliers and the management of the consequent contracts throughout their lifecycles. Each of these tools, however, is aimed at the OEM side of the relationship, while the primary tools used on the CM side are spreadsheet based and developed in-house. This highlights the opportunity in the market for software that enhances the existing RFQ process in relation to CM quotation response.

The major considerations for CMs when developing an RFQ response are product demand, supply constraints, lead time for a product through the manufacturing facility, raw material stock levels, purchasing constraints, and the lead time on components ordered. Unfortunately, the static nature of the software currently available to CMs does not permit the inclusion of the stochastic analysis required to fully investigate many of these factors. CMs have expressed a particular requirement for a means to calculate the amount of raw material they must hold in storage, in order to comply with the supply agreement with their customer, as this creates cost for the company in two ways. Firstly, there are the direct costs of storage space and preservation, and secondly, there are the costs associated with tying up large amounts of capital (such as lost interest and missed investment opportunities).

In an endeavour to broaden the scope of quote estimation to include cost drivers that are currently overlooked (see Fig. 2), discrete event simulation was nominated as a means of providing the required experimentation and analytical capabilities. To test the feasibility of this concept, a simulation model has been developed in association with the aforementioned field study companies. This model was designed to represent the pull of product through the system

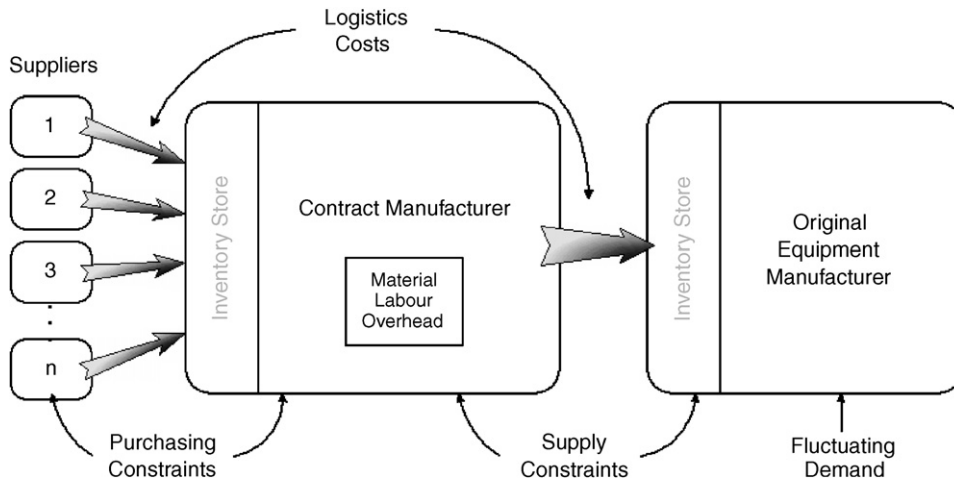


Fig. 2. Expanded scope of quote estimation.

by the OEM due to the potential outsourcing activity as per the RFQ and includes the major factors as outlined by the industrial representatives.

The model (described in detail in the next section) takes a number of inputs directly from incoming RFQs (annual demand volumes, bill of materials and order flexibility terms) and combines them with user inputs (such as component costs, lead times and production capacity) to perform the cost analysis. The objective of this model is to analyse the risk associated with varying stock levels and changes in supply and purchasing agreements, and the consequential effects on product cost and ability to meet dynamic customer demand. Providing the user with a range of possible results, rather than the single figure of their current systems, gives them a greater insight into the risk involved (as will be explained in more detail in Section 5). This allows for more informed decisions to be made in the heat of negotiations.

4. Model development

The model (Fig. 3) was built using eM-Plant, an object-oriented discrete event simulation package developed by Tecnomatix (www.tecnomatix.com). The object-oriented nature of this package is particularly useful as it allows for suppliers to be easily added to or deleted from the model. In this modelling approach the basic supplier frame is only created once, then from this generic structure all others are derived. The ODBC (open database connectivity) capabilities of eM-Plant allow the generic structure to be automatically customised to suit each of the individual suppliers by retrieving specific information for each supplier directly from an external database. Although the model was developed using data from a number of CM it was then tailored to the requirements of one particular company for initial testing. However, the overall system modelled is typical of the majority of outsourcing situations as analysed in our field research.

The simulation sequence begins in the demand frame, which controls the generation of demand across the run. This frame takes an annual volume as supplied in the RFQ and divides it across the year according to a user-defined pattern, which can be based on either historical demand or a customer forecast. This demand is then communicated to the OEM frame where it is used to generate a continuous 13 period forecast for the CM. Figs. 4 and 5 show how this demand is translated back along the supply chain and how each node responds. The OEM frame also contains the information concerning flexibility constraints as stipulated in the RFQ. These constraints allow the OEM to revise order quantities by restricted amounts dependent on the immediacy of the respective due-date, which in turn requires the manufacturer to revise its production and purchasing decisions. The model allows for the inclusion of charges for both initially dealing with a received order and also for amending orders changed under the flexibility terms of the contract.

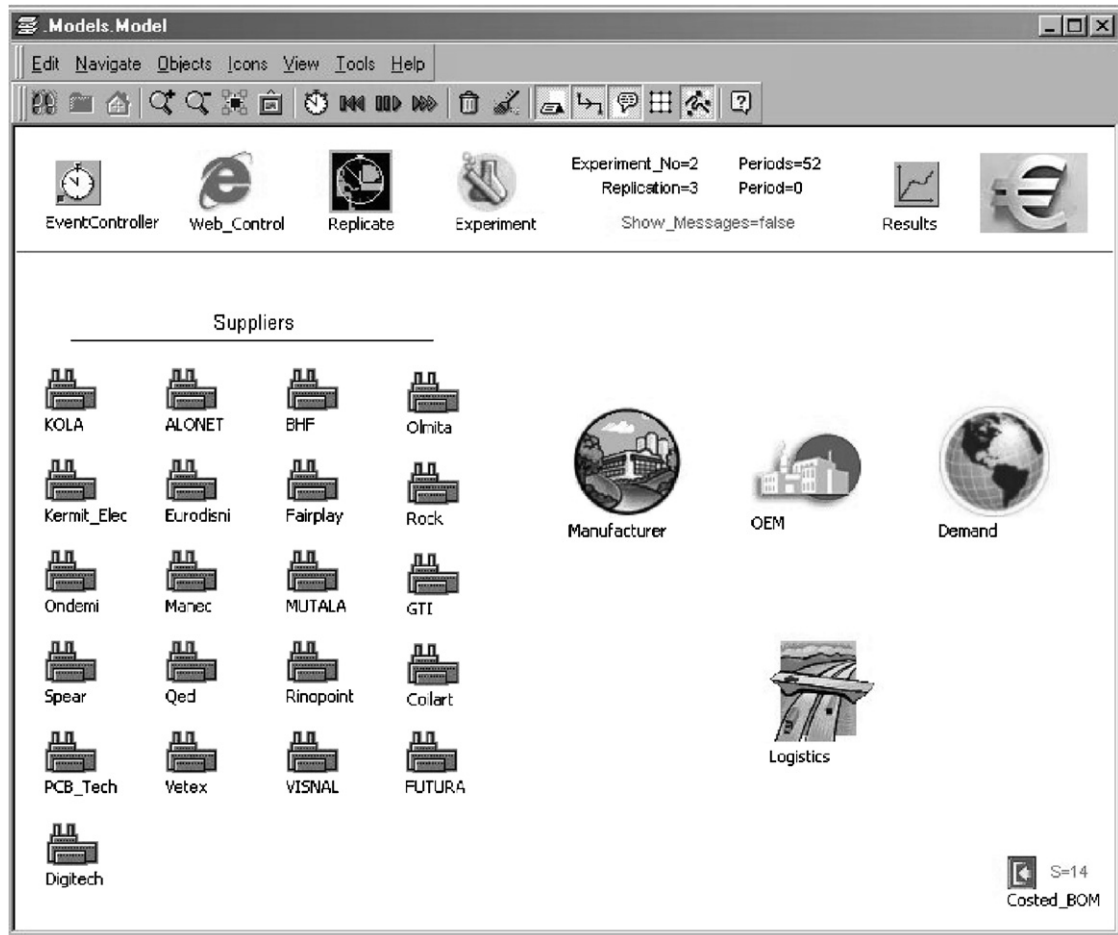


Fig. 3. Simulation model.

Within the manufacturer's frame, production capability is constrained by two factors. The first is production capacity, which is calculated from user inputs relating to setup time, processing time and overall time dedicated to the product in question. The second is a material constraint and relates directly to how many products can be made with the quantity of raw material in stock. The lower of these constraints determines how many products are manufactured in each period. If an order for a particular period is less than both these constraints, the model reviews the next four periods of the forecast for any potential bottlenecks and uses the extra capacity to help alleviate the problem. The extra product produced is then held in-house until required.

The reordering of material is based on both the forecasted demand and the lead time associated with that component. As there can be multiple potential suppliers for each component, the supplier of each component is selected at the initialisation stage of the simulation (this selection process is currently based on either lowest cost or shortest lead time). Once the stock level drops below the calculated reorder point the model relates the demand back to the corresponding supplier frame.

Each of the supplier nodes contains production details and cost information particular to that supplier, which is read directly from a database at the beginning of a simulation run. Once a supplier frame receives an order, the model satisfies as much of it as possible from its finished stock and places the remainder on back-order. The supplier then begins production in order to satisfy the backorder and replenish the finished stock to the desired level dependent on the stocking policy being used. After a delay based on the shipping time from the supplier the goods arrive at the manufacturing frame and the appropriate stock level figures are modified to reflect the additional raw material in the store.

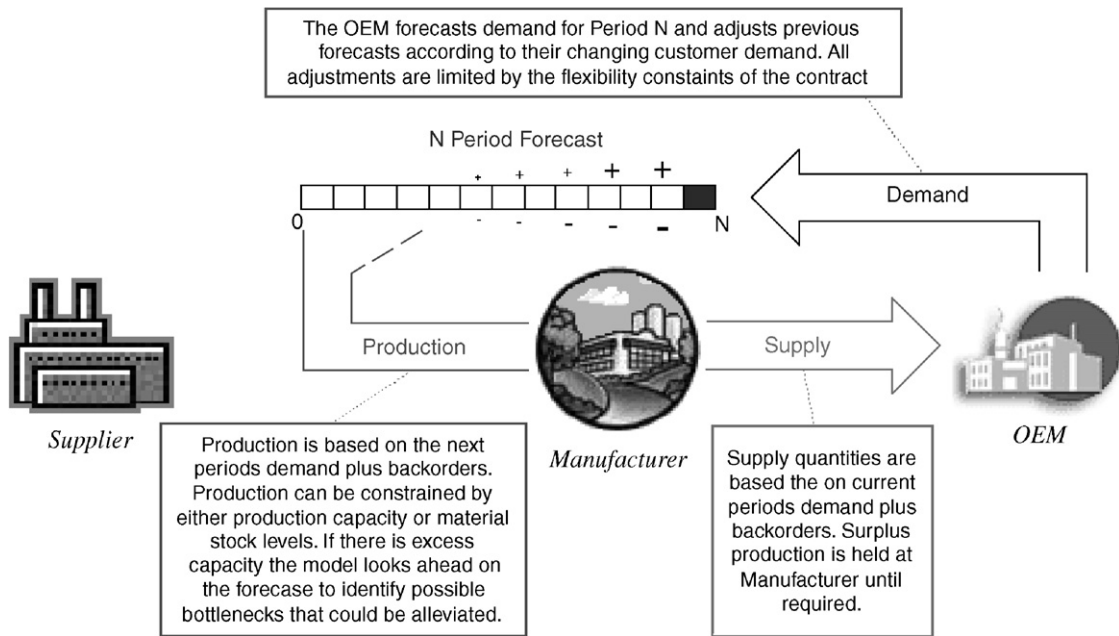


Fig. 4. Model interaction between manufacturer and OEM.

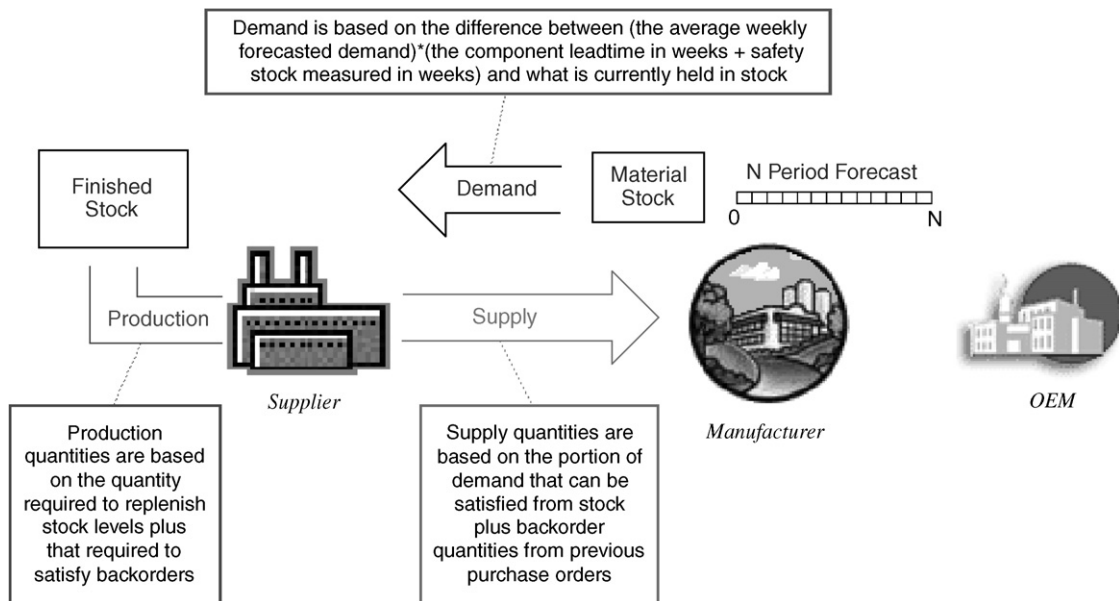


Fig. 5. Model interactions between the manufacturer and a supplier.

The current model has been set to simulate a time span of one year, which is segregated into one-week periods. These units have been chosen to correlate with the typical information received in an RFQ such as annual volumes and fixed period forecasts. Results from the model include a unit cost for the product, which is subdivided into material, logistics, storage, processing, administration and penalties incurred for late delivery. The model also records the OEM

demand for each period and the quantity that the manufacturer supplied, thus allowing a figure for on-time deliveries to be calculated.

5. Model experimentation overview

This section presents the general experimental capabilities of the model and describes how results are displayed. A more detailed analysis of a particular aspect of the experimentation capability is presented in Section 5.1. In the case study examined in this paper, *Company X* received an RFQ for the provision of PCBAs. This RFQ document contained amongst other information details of forecasted demand, a bill of materials, an approved vendor listing, board drawings and flexibility constraints. The aim of the model experimentation is to provide the user with greater insight into the implications for their system if the possible scenarios, as allowed in the terms of the proposed contract, were to be realised.

From demonstrating prototypes to representatives of *Company X* it has been noted that the range of values for possible results can prove most informative for the user as they indicate the level of risk associated with decisions. The simulation model is designed to replicate the system a number of times using different seed values for the generation of stochastic elements, thus producing a different snapshot of a possible reality each time. The results of each replication are then displayed graphically so that the extreme values can be quickly ascertained and the user is presented with an overall profile of possible outcomes. This differs somewhat from the standard approach to using simulation where one runs multiple replications and obtains the average and confidence interval. The decision to present the results in this manner was made in consultation with the field study companies, who showed preference to more graphical descriptive statistics rather than traditional mathematical statistics. It was felt that industrial users unaccustomed to classical statistics would be more receptive to a visual display that would convey information on spread and skewness intuitively.

Table 2 shows 11 different combinations of experimental variable settings that were selected for initial analysis. The first three of these variables allow the user to modify the demand pattern used in the model. The annual demand can be divided either evenly across the year or according to a known demand pattern (from forecast or historical records). The model permits this pattern to be altered randomly (“With spikes”) in order to study the effects of uncertainty in the forecast. The “Spike frequency” and “Spike factor” variables control the random rate of occurrence and scale of these demand alterations. The “Use forecast” option relates to whether or not the manufacturer frame of the model uses the forecast to alleviate bottlenecks. The “Allow flexibility” option allows the user to examine the effect of including the flexibility terms of the contract in their analysis. The “Flex exp” variable alters the behaviour of the contract flexibility, for instance the “Max_upward” experiment forces the model to increase the order quantities by the maximum amount each time it is triggered. Conversely, the “Max_downward” experiment decreases order quantities by the maximum amount each time. When the “Uniform” option is selected, the degree of the flexibility employed is randomly chosen

Table 2
Experiment settings

Experimental run no.	Demand	Spike freq.	Spike factor	Use forecast	Allow flexibility	Flex exp	Flex prob	Allow overtime
1	Level	0	0	True	False	Uniform	0.1	False
2	Pattern	0	0	True	False	Uniform	0.1	False
3	With spikes	0.3	0.9	True	False	Uniform	0.1	False
4	With spikes	0.9	0.3	True	False	Uniform	0.1	False
5	Pattern	0	0	True	True	Uniform	0.05	False
6	Pattern	0	0	True	True	Uniform	0.1	False
7	Pattern	0	0	True	True	Uniform	0.2	False
8	Pattern	0	0	True	True	Max_upward	0.1	False
9	Pattern	0	0	True	True	Max_downward	0.1	False
10	Pattern	0	0	True	True	Uniform	0.1	False
11	Pattern	0	0	True	True	Max_upward	0.1	False

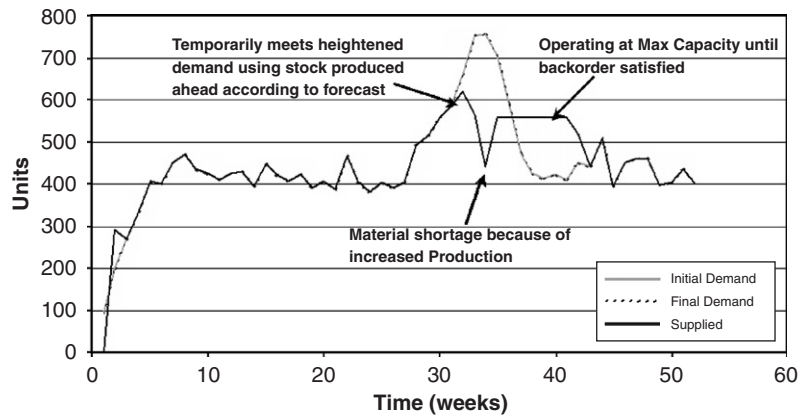


Fig. 6. Supply compliance for demand pattern.

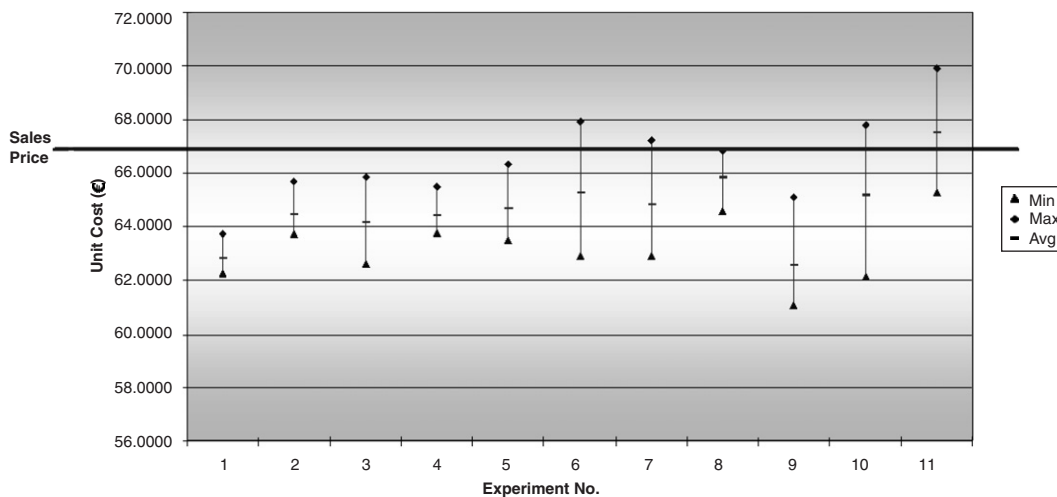


Fig. 7. Range of unit cost values for each experimental run.

between the constraint limits, as outlined in the RFQ, according to the uniform distribution. Meanwhile, the chance of the flexibility constraints being triggered is controlled by the “Flex prob” variable. The “Allow overtime” variable simply determines whether the model includes or excludes an overtime allowance in the calculations.

On completion of the simulation run, the model provides supply and demand information from across the simulated time span. For example, Fig. 6 shows the results of Experimental run no. 2, which excluded the flexibility terms that allow the OEM to change order quantities, as such initial and final demand figures coincide. In this graph one can see that during the time of elevated sales the model attempts to meet the increased demand using material that was built for stock in some of the earlier periods of less demand. However, the heightened production resulted in a drop in raw material stock levels, so the model is then constrained by the material capacity. Once the material stock has been replenished the manufacturer runs at maximum capacity until the backorder has been satisfied. At this point the manufacturer reverts back to supplying just the quantity as ordered by the OEM. As can be seen in Table 2, this experiment (No. 2) did not include overtime in the analysis. A resultant graph like this in Fig. 6 suggests that changes need to be made to both the production capacity (perhaps by including overtime) and the stock reordering strategy in order to cope with rapid increases in forecasted demand.

The graph in Fig. 7 shows the minimum, maximum and average values for unit cost over 15 replications for each experimental run in Table 2. It can be seen that as one includes the stochastic features of the contract the range on the

unit cost increases, this range essentially represents the risk involved in such a situation. For instance, if the case of level demand were to be true as in experimental run one then there is a minimal risk associated with setting the unit cost as the only significant sources of variation are related to incoming material. However, when one looks at the more realistic situation as modelled in experimental run six, where there is a fluctuating demand pattern and a changing forecast due to the inclusion of flexibility on order quantities, there is a much larger risk when setting the quote price. Therefore, in a price negotiation the user will be aware of when they are entering into a price range where they may occasionally produce at a loss and will be able to make a more educated decision when agreeing to a final price. The risk involved is due to the fact that if one sets the sales price too far above the upper end of the range, in the hope of covering all possibilities, they may not win the contract, whereas if they set the price too low they risk producing the product at a loss.

The experimentation discussed here relates only to a sample of the experiments made feasible through the use of discrete event simulation. Not only can the user experiment with the variables of Table 2 as discussed previously but also with any of the other inputs to the model. In this manner the user can evaluate a vast number of scenarios such as those with alternative suppliers or contract conditions. For the purposes of demonstration the next section focuses on one such experiment, which examines the flexibility terms of a proposed outsourcing contract.

5.1. Experiment demonstration—order flexibility terms

The simulation model presented in this paper provides a means of experimenting with the currently overlooked order flexibility terms. By analysing the risk associated with the prescribed terms and running experimental analysis on alternative terms, a contractor may be able to bring new options to the negotiating table (for example they may agree to a 3% decrease in sales price provided the flexibility terms are tightened by 10%).

The case here looks at the RFQ that *Company X* received for the provision of a PCBA. This RFQ contained a set of flexibility constraints shown here as Experiment 1 in Table 3. Experiment 2 shows the constraints used in the first alternative where the periods to due date were held the same and the flexibility terms were reduced. Experiment 3 investigated using the initial flexibility percentages but extending the timeline. Finally, Experiment 4 examined the combination of reducing the flexibility and also giving the contractor more notice of the changes.

Table 3
Order flexibility terms

Experiment	Periods to due date	Flexibility upwards (%)	Flexibility downwards (%)
1	2	0	0
	4	50	50
	8	75	75
	12	100	100
2	2	0	0
	4	20	20
	8	40	40
	12	100	100
3	4	0	0
	8	50	50
	12	75	75
	13	100	100
4	4	0	0
	8	20	20
	12	40	40
	13	100	100

After initially plotting simply the maximum, minimum and average values as seen previously in Fig. 7, it was agreed with the field study companies that Tukey's boxplots (sometimes called a box-and-whisker plots) [12] were more suitable for presenting the results. This is due to their ability to present a larger amount of data while retaining clarity. Boxplots have been noted to have the following strengths:

- Graphically display a variable's location and spread at a glance.
- Provide some indication of the data's symmetry and skewness.
- Unlike many other methods of data display, boxplots show outliers.
- By using a boxplot for each categorical variable side-by-side on the same graph, one can quickly compare data sets.

One acknowledged drawback of boxplots is that they tend to emphasise the tails of a distribution, which are the least certain points in the data set. Although in the situation studied here, each of the outliers is still a feasible outcome under the terms of the RFQ and may therefore still be of interest to the user when analysing risk. Fig. 8 shows the box plot results for unit cost for each of the experiments outlined in Table 3.

As can be seen from Fig. 8, reducing the percentage values of the flexibility constraints in Experiment 2 from 50% and 75% to 20% and 40%, respectively, reduced the cost of producing the PCBA. Alternatively, in Experiment 3 each of the times between the cut-off points and the due date values were increased while maintaining the same flexibility figures, which proved to have an even greater effect on reducing cost. This experiment also had a significant effect on the spread of the result variable, suggesting a reduction of volatility in the system and thus a more maintainable working environment. As expected, the combination of lower percentage flexibilities and increased time periods as used in Experiment 4 yielded the lowest median cost and least variability.

Fig. 9 presents the corresponding on-time delivery results for each experiment. One can see that in the vast majority of cases an on-time delivery rate of 100% is not achieved. This is explained by the fact that the demand forecast received by *Company X* contained periods of demand that exceeded their production capacity. As such further negotiations are required to resolve this issue. Possible solutions include increasing production capacity and charging a premium or supplying ahead of time if the OEM is willing to incur the added storage costs.

The results presented in Figs. 8 and 9 are based on 30 replications of the simulation model for each experiment. The individual results for each replication are provided in Appendix A. One can observe how the use of box plots would provide a user with a succinct understanding of the significant values while identifying outliers in an exceptionally short timeframe.

Initial results give reason to believe that there is merit in using simulation for CM contract costing. The insight gained through this analysis would allow a contractor to return to the OEM with a set of sales prices dependent on the

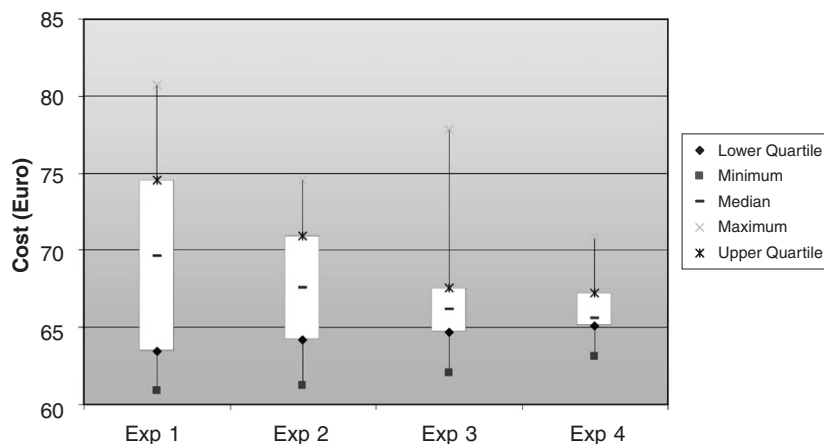


Fig. 8. Unit cost results.

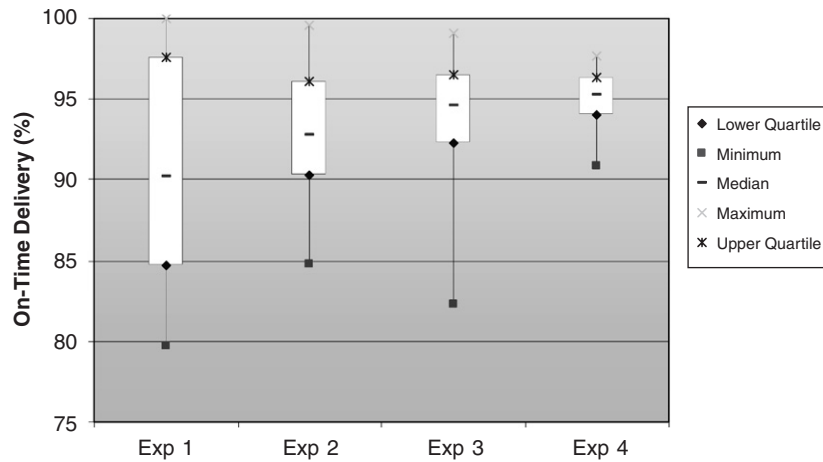


Fig. 9. On-time delivery results.

flexibility terms they are willing to accept. The fact that the contract has been subjected to this analysis places the CM in a much stronger position during the negotiation and can offer the OEM more confidence in the quoted prices.

6. Conclusions and future work

In this paper a deficiency in the outsourcing software market has been identified. Information obtained from a number of companies dealing directly with outsourcing in the electronics sector has highlighted a variety of requirements for which solutions have yet to be found in currently available software. Notably, at present there are specific elements of common contracts that respondents are regularly forced to overlook due to lack of analytical capabilities. Any means of analysing stipulations that are currently accepted a priori would open new avenues of negotiation and facilitate the abolition of contract constraints that unnecessarily increase the cost of providing a product/service.

Regardless of the scale of the company or of the contract, each company studied based their RFQ quotes on the three basic costs of material, labour and overhead. Initial research into possible solutions for the problems faced by CMs has focused on discrete event simulation. This approach has firstly been identified as a means of broadening the scope of quote calculation to incorporate elements such as logistics and, secondly, as an instrument for analysing stochastic elements of outsourcing contracts. It is proposed that greater analysis of RFQs can have benefits for both sides of the agreement; the CM will gain a better understanding of the implications inherent to the contract they are entering into and the OEM will experience a reduced risk of repeating the time-consuming RFQ process.

Initial results from both the field study and model development have been thoroughly encouraging and suggest that there is merit in using simulation for contract costing. Future work in this field includes more rigorous testing of the model in varying manufacturing environments and consequently refining the detail of the simulation. Further study is also necessary in regard to the execution of experiments and the presentation of the ensuing results, in order to concisely quantify for the end user the risk associated with an RFQ.

Making the benefits of such a system amenable to industrial personnel, who may be unfamiliar with discrete event simulation, is seen as crucial to its dissemination. For this reason, the long-term aspiration is to incorporate the elementary modelling technique into a functional tool that will provide the user with a level of abstraction from the complexities of simulation.

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Appendix A.

Below is shown the table of results per replication from experiment conducted in Section 5.1.

Replication	Experiment 1		Experiment 2		Experiment 3		Experiment 4	
	cost (€)	On-time (%)	cost (€)	On-time (%)	cost (€)	On-time (%)	cost (€)	On-time (%)
Rep no. 1	69.9396	87.4882	63.7466	97.2373	66.7428	93.0205	65.3155	96.1378
Rep no. 2	64.2586	99.2655	71.4807	89.1353	67.8722	93.4360	66.5907	94.3627
Rep no. 3	69.6481	92.1437	67.1642	91.4791	65.2205	93.8909	64.6132	97.1899
Rep no. 4	71.4346	87.8101	66.7374	93.8925	64.6764	96.4827	70.4203	91.7765
Rep no. 5	69.8703	88.3283	66.5475	92.7346	66.3115	94.5716	67.2684	93.2979
Rep no. 6	64.1695	96.5460	70.8779	87.8229	68.1395	92.2942	65.1997	94.5262
Rep no. 7	80.4247	81.8960	71.6539	87.3277	66.3712	92.5736	66.7128	94.5266
Rep no. 8	61.3150	94.5773	66.2312	95.4830	62.7857	98.0582	65.2511	96.1380
Rep no. 9	61.1177	99.9544	63.3153	97.0499	65.1025	94.5698	68.6545	91.2979
Rep no. 10	73.0259	86.4111	62.0955	98.8060	65.9021	96.2658	66.1192	95.2962
Rep no. 11	74.7543	86.4988	74.5402	84.7703	70.9745	89.3231	65.5372	96.6416
Rep no. 12	62.0897	98.8303	64.2184	96.1300	62.8684	98.6123	65.3172	96.6689
Rep no. 13	61.2476	99.3488	62.1484	98.0962	68.9479	92.1986	64.9961	96.5441
Rep no. 14	79.8758	79.7484	69.1468	91.0127	68.3836	91.7602	64.8664	95.2997
Rep no. 15	66.2826	96.4165	61.2557	99.2036	63.7489	95.7317	65.8761	95.4486
Rep no. 16	74.3829	86.3679	68.0333	93.1210	62.6252	99.0453	65.7158	95.2200
Rep no. 17	74.8078	84.3144	70.9170	90.3030	66.5612	94.4759	63.0984	97.6239
Rep no. 18	79.7117	80.5664	71.0009	90.7366	67.2905	94.1075	67.3048	94.9940
Rep no. 19	74.1182	84.6397	71.5515	88.4454	62.4519	97.2160	65.2572	96.3452
Rep no. 20	62.4846	98.6782	65.1788	97.7519	65.9131	95.1494	64.5341	95.7708
Rep no. 21	66.6003	93.4503	69.2773	92.8366	65.9536	94.5958	65.8014	95.6371
Rep no. 22	74.5595	84.7097	68.0436	92.6811	77.8802	82.3458	64.8781	95.6308
Repno. 23	63.3034	97.5761	63.8272	99.5741	66.9206	91.9397	66.1797	93.9418
Rep no. 24	80.7610	80.9478	63.2705	95.9526	62.0806	98.8205	65.0870	97.6435
Rep no. 25	74.5593	84.1970	66.0857	93.1945	66.4737	97.6637	68.2013	93.6339
Rep no. 26	68.0656	92.8997	72.6172	86.5376	72.6204	87.8889	67.2225	94.0546
Rep no. 27	64.5541	97.4666	68.3043	92.4286	64.7092	97.7402	65.5480	95.7631
Rep no. 28	60.9260	99.0328	69.0793	92.8362	97.5741	91.4023	70.8376	90.8956
Rep no. 29	73.7189	84.9754	72.8203	90.3176	64.7165	95.2182	69.2248	91.9034
Rep no. 30	63.4339	100.0000	66.5378	91.5872	63.7341	96.3628	64.1394	97.4411

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